

Data Base Lab
Lab Task 10



Malik Muhammad Sanaullah
24P-0554
Section 4A

Task 01

```
test> use 24p-0554
switched to db 24p-0554
24p-0554> |
```

Task 02

```
switched to db 24p-0554
24p-0554> db.createCollection("users")
{ ok: 1 }
24p-0554> db.createCollection("posts")
{ ok: 1 }
24p-0554> |
```

Task 03

```
24p-0554> show collections
posts
users
24p-0554> |
```

Task 04

```
24p-0554> db.users.insertOne({
|   name: "Muhammad Sanaullah",
|   email: "sanaullah@pwr.pk",
|   age: 20
| })
{
  acknowledged: true,
  insertedId: ObjectId('69efa276ea1ad312453682d1')
}
24p-0554> |
```

Task 05

```
24p-0554> db.users.insertMany([
|   {
|     name: "Sana Malik",
|     email: "sana@pwr.pk",
|     age: 21
|   },
|   {
|     name: "Saif Malik",
|     email: "saif@pwr.pk",
|     age: 18
|   },
|   {
|     name: "Sani Malik",
|     email: "sani@pwr.pk",
|     age: 22
|   },
|   {
|     name: "Malik Mustafa",
|     email: "malik@pwr.pk",
|     age: 23
|   }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69efa35dea1ad312453682d2'),
    '1': ObjectId('69efa35dea1ad312453682d3'),
    '2': ObjectId('69efa35dea1ad312453682d4'),
    '3': ObjectId('69efa35dea1ad312453682d5')
  }
}
24p-0554> |
```

Task 06

```
24p-0554> db.users.find()
[
  {
    _id: ObjectId('69efa276ea1ad312453682d1'),
    name: 'Muhammad Sanaullah',
    email: 'sanaullah@pwr.pk',
    age: 20
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d2'),
    name: 'Sana Malik',
    email: 'sana@pwr.pk',
    age: 21
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d3'),
    name: 'Saif Malik',
    email: 'saif@pwr.pk',
    age: 18
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d4'),
    name: 'Sani Malik',
    email: 'sani@pwr.pk',
    age: 22
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d5'),
    name: 'Malik Mustafa',
    email: 'malik@pwr.pk',
    age: 23
  }
]
24p-0554> |
```

```

24p-0554> db.posts.insertMany([
  {
    userId: ObjectId("69efa276ea1ad312453682d1"),
    content: "Student at Fast Peshawar",
    likes: 9999,
    createdAt: new Date()
  },
  {
    userId: ObjectId("69efa35dea1ad312453682d2"),
    content: "Student at NUST",
    Likes: 9499,
    createdAt: new Date()
  },
  {
    userId: ObjectId("69efa35dea1ad312453682d3"),
    content: "Currently at home doing nothing",
    likes: 8999,
    createAt: new Date()
  },
  {
    userId: ObjectId("69efa35dea1ad312453682d4"),
    content: "Looking for job",
    likes: 7999,
    createAt: new Date()
  },
  {
    userId: ObjectId("69efa35dea1ad312453682d5"),
    content: "Not born yet",
    likes: 7500,
    createAt: new Date()
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69efa722ea1ad312453682d6'),
    '1': ObjectId('69efa722ea1ad312453682d7'),
    '2': ObjectId('69efa722ea1ad312453682d8'),
    '3': ObjectId('69efa722ea1ad312453682d9'),
    '4': ObjectId('69efa722ea1ad312453682da')
  }
}
24p-0554> |

```

Task 07

```
24p-0554> load("C:/social_data.js")
Starting MongoDB Data Insertion Script

Inserting Users:
Users inserted successfully!
Total users: 11

Inserting Posts:
Posts inserted successfully!

Summary:
Total users in database: 11
Total posts in database: 15

Script execution completed!
true
24p-0554> |
```

Task 8

db.users.find().pretty()

```
24p-0554> db.users.find().pretty()
[
  {
    _id: ObjectId('69efa276ea1ad312453682d1'),
    name: 'Muhammad Sanaullah',
    email: 'sanaullah@pwr.pk',
    age: 20
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d2'),
    name: 'Sana Malik',
    email: 'sana@pwr.pk',
    age: 21
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d3'),
    name: 'Saif Malik',
    email: 'saif@pwr.pk',
    age: 18
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d4'),
    name: 'Sani Malik',
    email: 'sani@pwr.pk',
    age: 22
  },
  {
    _id: ObjectId('69efa35dea1ad312453682d5'),
    name: 'Malik Mustafa',
    email: 'malik@pwr.pk',
    age: 23
  },
]
```

```
db.posts.find().pretty()
```

```
24p-0554> db.posts.find().pretty()
[
  {
    _id: ObjectId('69efa722ea1ad312453682d6'),
    userId: ObjectId('69efa276ea1ad312453682d1'),
    content: 'Student at Fast Peshawar',
    likes: 9999,
    createdAt: ISODate('2026-04-27T18:12:50.633Z')
  },
  {
    _id: ObjectId('69efa722ea1ad312453682d7'),
    userId: ObjectId('69efa35dea1ad312453682d2'),
    content: 'Student at NUST',
    Likes: 9499,
    createdAt: ISODate('2026-04-27T18:12:50.633Z')
  },
  {
    _id: ObjectId('69efa722ea1ad312453682d8'),
    userId: ObjectId('69efa35dea1ad312453682d3'),
    content: 'Currently at home doing nothing',
    likes: 8999,
    createAt: ISODate('2026-04-27T18:12:50.633Z')
  },
  {
    _id: ObjectId('69efa722ea1ad312453682d9'),
    userId: ObjectId('69efa35dea1ad312453682d4'),
    content: 'Looking for job',
    likes: 7999,
    createAt: ISODate('2026-04-27T18:12:50.633Z')
  },
  {
    _id: ObjectId('69efa722ea1ad312453682da'),
    userId: ObjectId('69efa35dea1ad312453682d5'),
    content: 'Not born yet',
    likes: 7500,
    createAt: ISODate('2026-04-27T18:12:50.633Z')
  }
]
```



```
db.users.countDocuments()  
db.posts.countDocuments()
```

```
24p-0554> db.users.countDocuments()  
11  
24p-0554> db.posts.countDocuments()  
15  
24p-0554> |
```

Task 9

```
db.posts.drop()  
  
show collections
```

```
24p-0554> db.posts.drop()  
true  
24p-0554> show collections  
users  
24p-0554> |
```

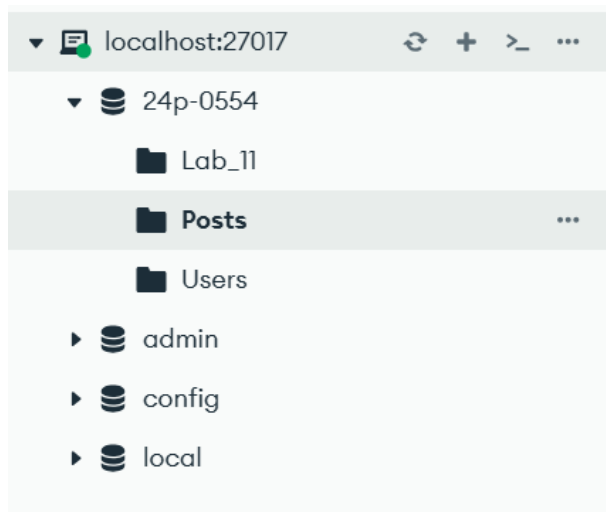
Task 10

```
24p-0554> db  
24p-0554  
24p-0554> db.dropDatabase()  
{ ok: 1, dropped: '24p-0554' }  
24p-0554> show dbs  
admin      40.00 KiB  
config     104.00 KiB  
local      64.00 KiB  
24p-0554> use 24p-0554  
already on db 24p-0554  
24p-0554> db.users.find()  
  
24p-0554> |
```

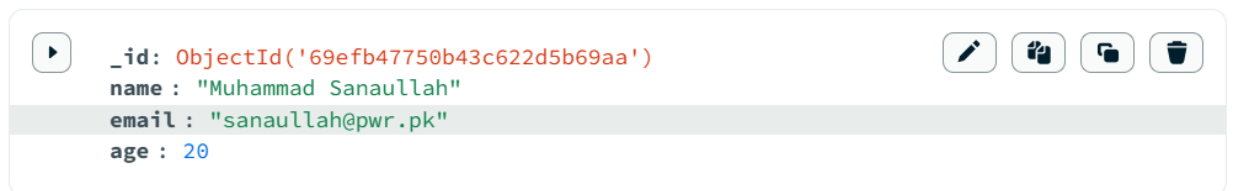
MongoDB Compass

Task Solutions

Task 1,2,3:



Task 04



Task 05

<

Task 06

```
_id: ObjectId('69efb58b50b43c622d5b69b2')
userId : "69efa276ealad312453682d1"
content : "Student at Fast Peshawar"
likes : 9999
createdAt : "2026-04-27T18:12:50.633Z"
```

```
_id: ObjectId('69efb58b50b43c622d5b69b3')
userId : "69efa35deald312453682d2"
content : "Student at NUST"
likes : 9499
createdAt : "2026-04-27T18:12:50.633Z"
```

```
_id: ObjectId('69efb58b50b43c622d5b69b4')
userId : "69efa35deald312453682d3"
content : "Currently at home doing nothing"
likes : 8999
createdAt : "2026-04-27T18:12:50.633Z"
```

```
_id: ObjectId('69efb58b50b43c622d5b69b5')
userId : "69efa35deald312453682d4"
content : "Looking for job"
likes : 7999
createdAt : "2026-04-27T18:12:50.633Z"
```

Task 08

The screenshot displays the MongoDB Compass application interface. On the left, a sidebar shows the database structure for 'localhost:27017', including a collection '24p-0554' with sub-collections 'Lab_11', 'Posts', and 'Users'. The main panel shows the 'Users' collection with five documents. Each document contains fields for '_id', 'name', 'email', and 'age'. The 'Posts' collection is also visible, showing five documents with fields for '_id', 'userId', 'content', 'likes', and 'createdAt'.

Collection	Document ID	Fields
Users	ObjectId('69efb47750b43c622d5b69aa')	name: "Muhammad Sanaulah", email: "sanaulah@pwr.pk", age: 20
	ObjectId('69efb4a550b43c622d5b69ac')	name: "Sana Malik", email: "sana@pwr.pk", age: 21
	ObjectId('69efb4a550b43c622d5b69ad')	name: "Saif Malik", email: "saif@pwr.pk", age: 18
	ObjectId('69efb4a550b43c622d5b69ae')	name: "Sani Malik", email: "sani@pwr.pk", age: 22
	ObjectId('69efb4a550b43c622d5b69af')	name: "Malik Mustafa", email: "malik@pwr.pk", age: 23
Posts	ObjectId('69efb58b50b43c622d5b69b2')	userId: "69efa276e1ad312453682d1", content: "Student at Fast Peshawar", likes: 9999, createdAt: "2026-04-27T18:12:50.633Z"
	ObjectId('69efb58b50b43c622d5b69b3')	userId: "69efa35de1ad312453682d2", content: "Student at NUST", likes: 9499, createdAt: "2026-04-27T18:12:50.633Z"
	ObjectId('69efb58b50b43c622d5b69b4')	userId: "69efa35de1ad312453682d3", content: "Currently at home doing nothing", likes: 8999, createdAt: "2026-04-27T18:12:50.633Z"
	ObjectId('69efb58b50b43c622d5b69b5')	userId: "69efa35de1ad312453682d4", content: "Looking for job", likes: 7999, createdAt: "2026-04-27T18:12:50.633Z"
	ObjectId('69efb58b50b43c622d5b69b6')	userId: "69efa35de1ad312453682d5", content: "Not born yet", likes: 7500, createdAt: "2026-04-27T18:12:50.633Z"

Task 09

The screenshot shows a notification banner at the top left stating 'Collection "24p-0554.Posts" dropped' with a green checkmark icon. The sidebar on the right shows the database structure for 'localhost:27017', including a collection '24p-0554' with sub-collections 'Lab_11' and 'Users'.

Task 10



Database "24p-0554" dropped

